

L02 Reflective Journal - ML Development Tools

Patrick Gandjouon Tchassem
ITAI 1371

This lab helped me understand how different Python tools work together in a machine learning workflow. Before working with the Iris dataset, I knew that pandas, NumPy, and matplotlib were important tools, but I did not fully understand how they connected. Working with the dataset made the purpose of each tool clearer to me. I saw how pandas can organize data into a DataFrame, NumPy can perform calculations, and matplotlib can turn the data into a visual form that is easier to understand.

One important thing I learned was that looking at raw data is not always enough to understand what is happening. When I first looked at the rows of the Iris dataset, I could see the measurements, but the relationships between the flower species were not obvious. The scatter plot made the differences easier to see. Setosa appeared clearly separated from the other species, while versicolor and virginica overlapped more. This helped me understand why visualization is an important step before building a machine learning model.

The statistical analysis also helped me connect numbers with what I saw in the visualization. The dataset had 150 samples, four features, and three classes, with 50 samples for each species. I learned that having the same number of samples for each class gives a balanced dataset. I also noticed differences in the average measurements between the species. This made me think more about which features might be most useful for classification. For example, petal length and petal width may separate the species better than using only sepal measurements.

Using NumPy for the mean and standard deviation helped me understand how numerical calculations can be performed quickly on a complete column of data. The mean sepal length was about 5.84 cm and the standard deviation was about 0.83 cm. At first, these calculations seemed like simple statistics, but I realized they are useful for understanding the center and spread of data before using it in a machine learning model.

Another lesson from this lab was the importance of checking code and outputs as I work. Running the notebook from beginning to end helped me verify that each cell depended correctly on the previous cells. I also learned that good documentation is part of the technical work. Comments, Markdown explanations, clear graphs, and meaningful variable names make a notebook easier for another person to understand and also make it easier for me to review my own work later.

Overall, this lab gave me a better understanding of the tools used before the actual machine learning model is built. My biggest takeaway is that exploring and understanding the data is an important part of machine learning, not just a preparation step to rush through. I would like to continue learning how these tools are used for data preprocessing and how different features affect the accuracy of a classification model.